DoangJoo "Alan" Synn

Ph.D. Student in Computer Science · Georgia Institute of Technology
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Research Interests

My research builds high-efficiency and scalable machine learning systems, with a focus on automated dataloader tuning, memory-aware and distributed training, and privacy-preserving inference. I work across cloud and distributed computing to make ML training and deployment faster, more resource-efficient, and easier to use.

Education

Georgia Tech, Ph.D. in Computer Science *2022–Present*
- School of Interactive Computing; advised by Profs. Oh & Ha
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

Korea University, M.Eng. in ECE *2020–2022*
- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

Korea University, B.Eng. in EE *2012–2020*

Experience

Ph.D. Research Intern, Dolby Laboratories *2024–2025*
- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

MLOps Engineering Intern, LG Uplus Corp. *2023*
- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

Research Scientist, Protopia AI Inc. *2021–2022*
- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.

Cloud Architect · AWS Educate Cloud Ambassador *2020–2022*
- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.

Server Engineer, Flysher Inc. *2019*
- Shipped backend features for four social-game titles; one reached Facebook's global top 10.

System Engineer, AKA Intelligence (Musio) *2017–2019*
- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.

Selected Publications

1 **D. Synn**, X. Piao, J. Park, and J. Kim, “Distributed Training of Deep Learning Models Using Micro Batch Streaming,” *KTSDE*, Mar. 2023.
2 **D. Synn** and J. Kim, “Num Worker Tuner: An Automated Spawn Parameter Tuner for Multi-Processing DataLoaders,” *ACK*, Oct. 2021.
3 B. Ahn, **D. Synn**, M. Derkani, E. Ebrahimi, and H. Esmaeilzadeh, “Protopia AI: Taking on the Missing Link in AI Privacy and Data Protection,” *NeurIPS Demo*, 2021.